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Hydroclimatic and flood inundation modelling of Lake Victoria using the LaVIWaB model

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ABSTRACT

Study region: The Lake Victoria Basin has experienced increasing shoreline flooding in recent decades due to interactions among hydroclimatic variability, land use change, and regulated outflows at the Nalubaale–Kiira (Owen Falls) dam complex. Rising lake levels and extreme rainfall events have intensified flood risks for low-lying littoral communities across East Africa. These trends highlight the need for basin-scale hydrological modelling to better understand lake dynamics and shoreline flooding.

Study focus: This study develops the Lake Victoria Integrated Water Balance (LaVIWaB) Model, a physically based framework designed to simulate lake level dynamics and flood inundation under varying climatic conditions. The model integrates multi-decadal observations (1984–2023) of precipitation, evaporation, river inflows, regulated outflows, and lake bathymetry using satellite-derived and in-situ datasets. Model calibration shows strong performance in reproducing observed lake levels (NSE = 0.749, R^2 = 0.749, RMSE = 0.262 m). Validation against satellite-derived lake surface area datasets and high-resolution DEM products shows moderate agreement in capturing lake extent variability (Cor = 0.598, RMSE = 22.77 km^2).

New hydrological insights for the region: The model reproduces seasonal and interannual variability in lake storage and better constrains the nonlinear lake level–area relationship. Once lake levels exceed ~1132 m a.s.l., small increases trigger disproportionate expansion of inundated areas across low-gradient littoral floodplains, highlighting escalating flood risks and the value of integrated water balance–inundation modelling for regional flood management.

1. Introduction

Floods are among the most devastating natural disasters, particularly in regions where large water bodies and extreme climatic conditions interact to amplify hydrological variability. In East Africa, Lake Victoria, the world's second-largest freshwater lake by

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surface area, has experienced an increasing frequency of extreme lake flooding events in recent years, resulting in significant socio-economic and environmental consequences (Nassali et al., 2020). These flood events have caused widespread displacement, destruction of infrastructure, loss of livelihoods, and disruption of critical economic sectors including agriculture, fisheries, hydropower generation, and transportation (Ogembo and Akinyi Mohamed, 2023). Understanding and predicting such hydrological extremes requires robust modelling frameworks capable of integrating multiple hydrological processes and climate variability drivers (Banda et al., 2022).

A process-based understanding of Lake Victoria's hydrology is essential to contextualize flood dynamics beyond individual events. The lake's water balance is primarily governed by direct precipitation over the lake surface, evaporation, tributary inflows, and regulated outflows, with precipitation contributing up to 80% of total inflow (Sutcliffe, 1999; Vanderkelen et al., 2018). Variability in these components operates across seasonal to decadal timescales and is strongly influenced by large-scale climate modes such as ENSO and the Indian Ocean Dipole (Nicholson, 2017a). Importantly, the translation of water level changes into flood extent is mediated by basin morphology, particularly the nonlinear hypsometric relationship between lake level, area, and volume (Messager et al., 2016).

While recent extreme events, particularly the 2019–2020 flood episode, illustrate the societal impacts of lake level rise, they are best interpreted within the broader context of hydroclimatic variability and basin-scale water balance dynamics (Khaki and Awange, 2021; Pietroiusti et al., 2024; Shinhu et al., 2023; Shu et al., 2024). Over the past decade, Lake Victoria has exhibited pronounced water level fluctuations driven by interactions between climatic variability and anthropogenic controls, including land-use change and regulated outflows (Shinhu et al., 2023). The 2019–2020 event, during which lake levels rose by ~1.4 m following anomalously intense MAM and OND rainfall, represents an extreme manifestation of these coupled processes (Khaki and Awange, 2021; Pietroiusti et al., 2024). Similar conditions in subsequent years (2024–2025) further highlight the increasing frequency of such hydroclimatic extremes. Collectively, these observations point to a system characterized by growing variability and sensitivity to both climatic forcing and human regulation (Ali et al., 2024a; Ondolo and Osio, 2025).

A growing body of literature has examined the drivers of Lake Victoria's water level variability and associated flood dynamics. Previous studies highlight the dominant role of precipitation–evaporation imbalances and tributary inflows in controlling lake storage, modulated by regulated outflows at the Nalubaale–Kiira dam complex (Kikoyo et al., 2023; Manirakiza et al., 2022). In addition, geomorphological controls, particularly low-gradient littoral zones, have been shown to amplify flood exposure by translating small vertical water level changes into substantial lateral inundation (Mugume et al., 2024). Attribution studies further emphasize the combined influence of natural climate variability and anthropogenic warming in intensifying extreme lake-level events (Kipyegon Bosuben et al., 2023; Pietroiusti et al., 2024). Collectively, these findings underscore the need for modelling frameworks that explicitly couple water balance processes with spatially dynamic inundation responses.

A key but often underrepresented aspect of large lake systems is the nonlinear response of surface area and inundation extent to changes in water level. In shallow lakes such as Lake Victoria, small vertical increases in water level can translate into disproportionately large lateral expansions of inundated area due to low-gradient littoral topography (J. L. Awange et al., 2019; Messager et al., 2016; Plisnier et al., 2023). This hypsometric control introduces threshold behaviour in flood dynamics, whereby incremental storage increases can trigger rapid shoreline expansion once critical elevation thresholds are exceeded. Capturing this nonlinearity is therefore essential for realistic flood modelling and risk assessment.

Although expanding in situ hydro-meteorological monitoring across the basin remains essential, observational densification alone cannot fully resolve Lake Victoria's complex hydroclimatic dynamics. The basin is transboundary, with spatially uneven and temporally inconsistent gauge networks that limit comprehensive long-term assessment. Moreover, lake flooding is governed not only by rainfall–runoff processes but by an integrated water balance involving precipitation, evaporation, tributary inflows, regulated outflows at Nalubaale–Kiira, and nonlinear bathymetric controls on lake level–area–volume relationships.

Despite advances in lake water balance modelling, most existing approaches treat lake surface area as static, thereby decoupling hydrological fluxes from spatial inundation dynamics. This simplification limits the ability to represent feedbacks between lake storage, shoreline expansion, and floodplain connectivity, particularly in morphologically complex systems such as Lake Victoria (Schumann et al., 2023; Vanderkelen et al., 2018). Bridging this gap requires integrating hydrological mass balance with dynamic representations of lake geometry, enabling direct translation of water level changes into spatially explicit flood dynamics.

To address these limitations, this study develops the Lake Victoria Integrated Water Balance (LaVIWaB) Model, which explicitly couples the mass balance equation with a high-resolution elevation–area relationship derived from integrated DEM–bathymetry datasets. This formulation enables dynamic simulation of inundated surface area as a function of lake level, thereby resolving the nonlinear floodplain response characteristic of Lake Victoria's low-gradient littoral zones. In contrast to earlier lumped formulations, LaVIWaB translates hydrological fluxes directly into spatially explicit flood extent, allowing quantitative identification of threshold-driven shoreline inundation and flood-prone zones under extreme hydroclimatic and regulated flow conditions. This integration of water balance closure with dynamic flood extent modelling represents the core methodological advancement of the framework. Accordingly, this study develops and evaluates a robust Lake Victoria Integrated Water Balance Model specifically tailored to simulate lake level dynamics and shoreline flood events under extreme hydrometeorological conditions in the Lake Victoria Basin.

2. Study area

Lake Victoria is the world's second largest freshwater lake with an area of approximately 68,800 km^2 (Fig. 1). It lies at an altitude of 1134 m above sea level (m a.s.l). It is a shallow freshwater system with a maximum depth of 83 m. Its basin sustains over 42 million people and supports significant inland fisheries producing approximately 1 million ton of fish per year (Nyamweya et al., 2023). The water balance of the lake is mostly controlled through precipitation and evaporation: the lake basin receives an average annual rainfall

of 1400–2600 mm with evaporation rates of 1400–2400 mm (Ogembo and Akinyi Mohamed, 2023; Vanderkelen et al., 2018a). Under long-term mean conditions, direct precipitation contributes approximately 75–85% of total inflow to the lake, while river inflows account for the remaining 15–25% (Sutcliffe, 1999; Vanderkelen et al., 2018). Mean annual outflow at Jinja is on the order of ~30–40 $km^3\ yr^{-1}$, regulated by the Nalubaale–Kiira dam complex, linking lake levels directly to controlled discharge into the White Nile. The average depth of the lake is around 40 m while the maximum depth is around 80 m, hence Lake Victoria is considered to be a shallow lake.

3. Data and methods

3.1. Data

A suite of hydroclimatic datasets was compiled to drive and evaluate the Lake Victoria Integrated Water Balance Model (LaVIWaB). Table 1 provides an overview of the datasets, including sources, spatial/temporal resolution, and time periods. Of the employed datasets, seven serve as input to the LaVIWaB model, while three act as independent model evaluation datasets.

3.2. Original water balance model

The original water balance model (WBM) for Lake Victoria was introduced by Vanderkelen et al. (2018a) and later translated into Python by Pietroiusti et al. (2024). The model simulates lake storage and levels through a daily water balance equation Eq. (1):

$$\Delta L(t) = \frac{1}{A_f}\{P(t) - E(t) + Q_{in}(t) - Q_{out}(t)\} \quad (1)$$

where: $\Delta L(t)$= change in lake level (m), A_f = assumed constant lake surface area (m^2), P(t) = precipitation directly onto lake surface ($m^3\ day^{-1}$), E(t) = lake evaporation ($m^3\ day^{-1}$), $Q_{in}(t)$ = river inflows ($m^3\ day^{-1}$), and $Q_{out}(t)$ = outflows at Nalubaale/Jinja ($m^3\ day^{-1}$).

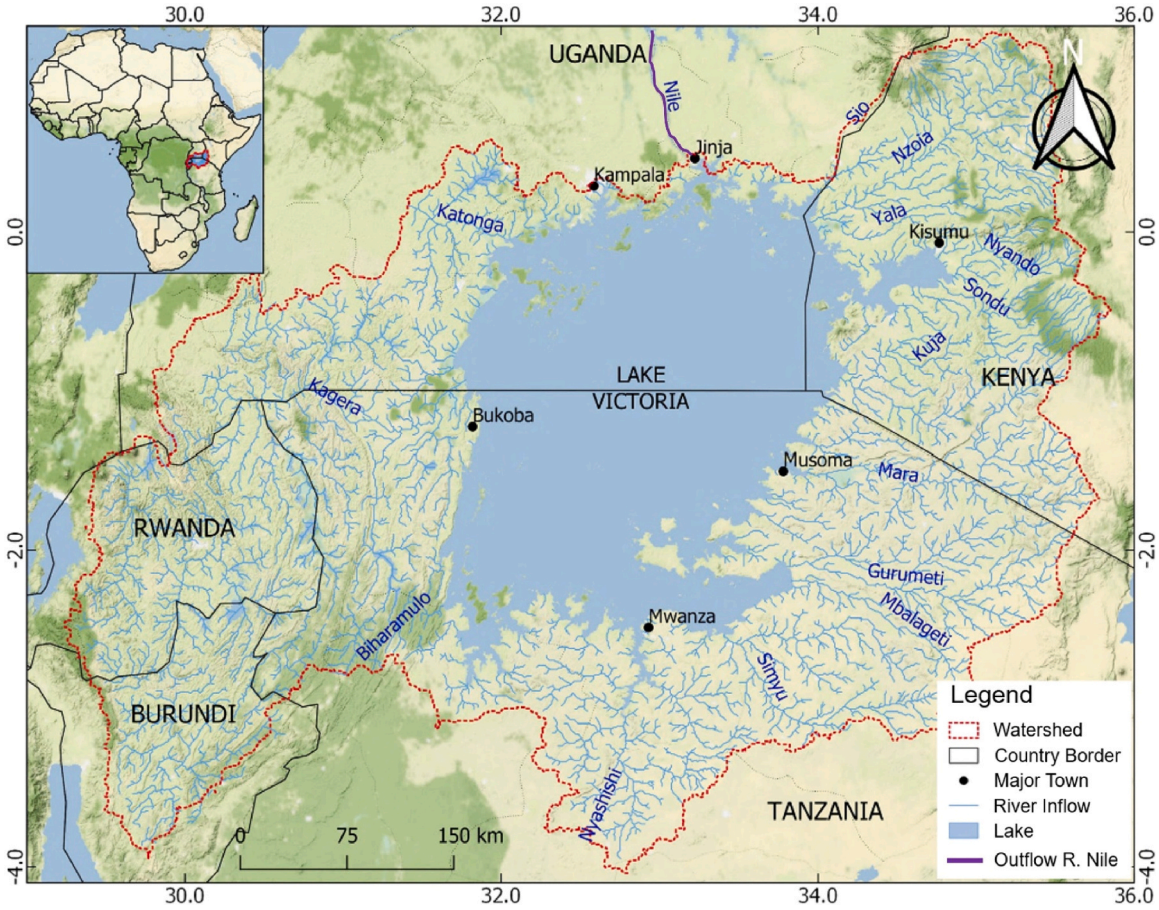


Fig. 1. Map of the Lake Victoria basin, showing major inflow tributaries across bordering countries as well as outflow into River Nile at Junja, Uganda. The green background is the topography of the basin. The inset shows the location of the basin in Africa.

Table 1
Summary of datasets used in this study.

Variable	Dataset/Source	Resolution	Period	Notes
Precipitation	PERSIANN-CDR (Ashouri et al., 2015, (Ogembo et al., 2026))	0.25° (~28 km), daily	1984–2023	Satellite-derived
Evaporation	COSMO-CLM^2 RCM forced with ERA5 (Thiery et al., 2015)	0.0625° (~7 km) daily	1981–2023	This is a climatology converted from latent heat flux (LHF)
Lake Outflow	Nalubaale Dam records, Agreed Curve (Sene, 2000; Vanderkelen et al., 2018a)	Daily	1948–2023	Hybrid series; regression corrections for the missing 2006–2011 values
Bathymetry and Topography	SRTM DEM (30 m), GEBCO bathymetry (500 m) (2023)	30 m	1984–2024	DEM + Bathymetry interpolated to 30 m resolution
Land Cover	GLAD Global Land Cover (2020)	30 m	2020	Used for Curve Number inflow calculation
Lake Levels (Evaluation)	WMO Hydromet (1948–1996), DAHITI (1993–2024)	Daily	1948–2024	DAHITI bias-corrected against in-situ data measured at Jinja, Uganda
Lake Area: time series (Evaluation)	Wu et al. (2023a), (2023b)	30 m	2020–2024	Observed extent from Wu et al. (2023a), (2023b)
Lake Area: Single points (Evaluation)	Copdem (2011); Dataverse (2015); WBM Vanderkelen et al. (2018a)	Global 500 m	2011, 2015, 2018	Independent model-based validation datasets of global lakes and basins

In the extended LaVIWaB framework, lake level $L(t)$is treated as the primary prognostic state variable, while lake surface area $A(t)$is treated as a diagnostic variable derived from lake level through a predefined hypsometric (area–elevation) relationship. Consequently, Eq. (1) governs the temporal evolution of lake level, and lake area is updated at each timestep as a function of the simulated lake level, i.e. $A(t) = A(L(t))$. The model was validated against observed levels from Hydromet and DAHITI. Hereafter we briefly describe the individual terms of the WBM as represented in the original model version; for further details we refer to Vanderkelen et al. (2018a) and Pietroiusti et al. (2024).

3.2.1. *Lake input terms*

Lake input was calculated as the sum of direct precipitation over the lake surface and riverine runoff from the surrounding basin. Over-lake precipitation was obtained from PERSIANN-CDR daily gridded data. For each timestep, the precipitation grid was masked using the corresponding lake surface extent, so that only grid cells falling within the lake contributed to the estimate. The precipitation values (in mm/day) for all lake water grid cells were then multiplied by their respective cell areas and summed to obtain the total volumetric input from direct rainfall over the lake surface. PERSIANN-CDR has proven reliable in providing precipitation estimates over the African continent, particularly in equatorial eastern Africa, where it outperforms other available products (Mekonnen et al., 2023). Prior to integration into the LaVIWaB framework, PERSIANN-CDR precipitation was bias-corrected against in situ gauge observations from the WMO Hydromet network over the Lake Victoria Basin. Monthly correction factors were computed over the overlapping period by deriving multiplicative bias ratios between spatially averaged gauge precipitation and corresponding satellite estimates. These correction factors were applied to the daily PERSIANN-CDR series to preserve temporal variability while adjusting systematic biases in magnitude. Residual uncertainty associated with satellite-based precipitation products was indirectly evaluated through model performance in reproducing observed lake levels and surface area dynamics. The propagation of precipitation uncertainty into flood extent simulations is therefore reflected in reported validation metrics including Nash–Sutcliffe Efficiency (NSE), Root Mean Square Error (RMSE), and Pearson correlation coefficient (r), providing an integrated assessment of hydrological realism.

Inflow from the basin is estimated with the Soil Conservation Service Curve Number (SCS-CN) method, widely used in hydrological modelling (Vanderkelen et al., 2018). Runoff depth is computed following Eq. (2) (Liu et al., 2022):

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)} \tag{2}$$

where: Q = Runoff depth (in inches), P = Precipitation depth (in inches), S = Potential maximum retention after runoff begins (in inches).

The Curve Number (CN) is related to S following Eq. (3):

$$S = \frac{25400}{CN} - 254 \tag{3}$$

Curve Number (CN) values were derived by integrating GLAD land cover classes with Hydrologic Soil Groups (HSG A–D) using standard SCS lookup tables. Spatial CN grids were generated by assigning CN values to each land cover–soil combination under antecedent moisture condition II (AMC-II), followed by area-weighted aggregation at sub-catchment scale.

Although discharge records exist for some major tributaries of Lake Victoria, their spatial and temporal coverage is incomplete across the transboundary basin, with data gaps and rating uncertainties affecting long-term consistency. Direct use of observed flows would therefore introduce spatial bias and discontinuities into the 1948–2024 simulation. The SCS-Curve Number method was adopted to provide a spatially consistent, basin-wide runoff estimate integrating land cover and precipitation across all sub-catchments. Given that the model targets lake-scale storage dynamics rather than detailed channel hydraulics, a parsimonious and

spatially distributed runoff approach is appropriate for this application.

Land cover classifications are sourced from the Global Land Cover and Land Use Change Dataset (GLAD) (2000–2020) which quantifies changes in forest extent and height, cropland, built-up lands, surface water, and perennial snow and ice extent from 2000 to 2020 at a 30-meter spatial resolution (Ogembo et al., 2022). This was an addition to the existing model for improved performance. Although locally derived land use classifications from Landsat imagery could provide site-specific validation, the GLAD Global Land Cover dataset was selected due to its consistent basin-wide coverage, standardized classification methodology, and documented performance across sub-Saharan Africa (Potapov et al., 2022).

3.2.2. Lake output terms

Where available, observational lake outflow at Jinja is used from Pietroiusti et al. (2024) and Vanderkelen et al. (2018a). Where not available, the Agreed Curve is used, a mathematical expression which relates observed lake levels to allowed discharge at the Nalubaale Dam in Jinja (Uganda; Fig. 1). Mathematically, the Agreed Curve is expressed following Eq. (4) (Sene, 2000):

$$Q_{out} = 66.3(L - 7.96)^{2.01} \quad (4)$$

Where Q_{out} is outflow in Jinja ($m^3\ s^{-1}$) and L indicates in-situ lake levels (m). Unlike previous applications in Pietroiusti et al., (2024) and Vanderkelen et al., (2018a) for the period between 5 March 2006 and 31 December 2011, when it is known that dam releases deviated from this regulation, outflow was reconstructed from a linear regression model between observed lake level and outflows in the period, as this was found to lead to better model performance (Appendix 3). Details of the regression formulation, fitted parameters, and validation metrics used for outflow reconstruction are provided in Appendix 3. We note, however, that for this ~6-year period the lake levels cannot be considered as a fully independent model evaluation product (see sect. 3.4).

Evaporation from the surface of Lake Victoria is calculated using the latent heat flux (LHF) from a COSMO-CLM^2 regional climate model simulation, in which it is forced by ERA-Interim reanalysis data (Thiery et al., 2015). The LHF is converted into evaporated water by dividing the flux by the latent heat of vaporization at 0 °C (2.5×10^6 J kg^{-1}). A yearly climatology of evaporation is derived by averaging daily evaporation values across years, which is then applied to all simulation years. Due to the challenges in accurately measuring evaporation over large water bodies, previous studies suggest that evaporation remains one of the largest sources of uncertainty in the water balance (Sutcliffe, 1999; Swenson and Wahr, 2009).

3.3. Model extension for lake inundated areas

To extend the original WBM, we incorporated a shoreline inundation module that links water levels to lake extent using DEM and bathymetry. This extension enables the dynamic simulation of inundated areas, critical for flood risk mapping. In this approach, each grid cell is treated as a binary state, either inundated or non-inundated, based on whether the simulated water level exceeds the cell's elevation. The instantaneous lake extent was then computed as shown in Eq. (5):

$$A(t) = \sum_{i=1}^{n} (a_i) \text{for all } hi \quad \leq L(t) \quad (5)$$

Where $A(t)$ is the lake surface area at time t (m^2), h_i is the elevation of pixel i, and a_i is the pixel area (900 m^2 for 30 m resolution).

The equation defines the diagnostic relationship between lake level and surface area derived from the DEM–bathymetry dataset. A (t) is not independently solved as a prognostic variable, but is directly computed from L(t) at each timestep. The temporal evolution of lake area is therefore implicitly governed by changes in lake level through the hypsometric relationship $A(L)$. The elevation–area (hypsometric) relationship $A(L)$used in the model is derived from the integrated DEM–bathymetry dataset. As a result, shoreline dynamics and inundation extent emerge directly from the simulated lake level without requiring a separate prognostic equation for area. The model follows a sequential update scheme: (i) lake level is updated using the water balance equation, and (ii) lake area and inundation extent are recomputed from the updated lake level using the DEM–bathymetry relationship.

Unlike previous Lake Victoria water balance applications that assume a constant lake surface area (Pietroiusti et al., 2024; Vanderkelen et al., 2018), the LaVIWaB extension dynamically updates lake geometry at each timestep. The model computes instantaneous inundation by comparing simulated lake levels to a high-resolution DEM–bathymetry composite surface, thereby resolving spatially explicit shoreline migration. This coupling of mass balance closure with dynamic area–elevation relationships enables nonlinear floodplain expansion to be captured within a single modelling framework.

3.4. Simulation set-up and evaluation

The LaVIWaB model was implemented in Python 3.11. The model was initialized using Hydromet-derived storage volumes from 1948, followed by a 3-year spin-up to equilibrate storage with meteorological forcing. Daily integration of the water balance calculation (Eq. 1–6) was performed for 1948–2023 to capture long-term lake dynamics. Inundation mapping was conducted by overlaying simulated water levels on a 30 m resolution DEM combined with bathymetric data to delineate shoreline positions and flooded areas. The model operates at a daily temporal resolution, consistent with the hydrometeorological forcing datasets used in the water balance calculations. Spatial inundation mapping is derived from a 30 m resolution DEM–bathymetry composite surface, meaning that simulated lake extent and flood-prone areas are represented at 30 m spatial resolution.

We evaluate our model simulations against three independent observational products (Table 1). First, simulated lake levels were evaluated against DAHITI satellite altimetry and WMO Hydromet in situ observations. The WMO Hydromet measurements are

available for the period 1948–1996, whereas DAHITI data is available for the period 1993–2024. Both products show high agreement in the overlapping time window (1993–1996), highlighting thus applicability as reference product for lake level evaluation (Vanderkelen et al., 2018).

Second, to evaluate the model-derived lake surface areas, we used observed lake extent data from Wu et al., (2023a), (2023b), which provides a time series of satellite-derived estimates of Lake Victoria's surface areas from 2000 to 2022. The dataset integrates Landsat imagery with advanced water classification algorithms and cross-validation against independent optical and radar records. For Lake Victoria, this dataset offers valuable empirical reference points for evaluating simulated inundation dynamics and shoreline variability. Observed absolute lake areas as well as anomalies from Wu et al., (2023a), (2023b) were extracted and compared against model-predicted extents derived from Eq. (6).

Third, the modelled lake extents were further validated against three snapshot datasets (COPDEM, 2011; Dataverse, 2015; Vanderkelen et al., 2018a; see Table 1) to assess the reliability of simulated areas across discrete validation years. The WBM datasets, developed using hydrologically conditioned DEMs and global water masks (Verpoorter et al., 2014; Yamazaki et al., 2019), provide consistent, hydro-topographically constrained basin geometries. Their cross-comparison with LaVIWaB outputs enables independent evaluation of the area–elevation relationships implemented in the model and long-term stability of the modelled lake surface representation. However, the lack of consistent long-term observations across the full simulation period limits their direct application an evaluation of temporal dynamics. Nonetheless, these products offer useful reference points for evaluating model performance in recent decades, thereby complementing the Wu et al., (2023a), (2023b) estimates. Model accuracy was assessed using the Pearson's correlation coefficient (r) and Root Mean Square Error (RMSE), and Bias, capturing both precision and systematic deviation.

4. Results

4.1. Hydroclimatic conditions and observations

Hydroclimatic analysis over the Lake Victoria Basin (1984–2023) reveals distinct spatial and temporal variability in key water balance components. Observed mean annual precipitation from PERSIANN-CDR for the period 1984–2023 highlights heterogeneous rainfall distribution influenced by topography and regional climate dynamics (Thiery et al., 2015, 2017; Fig. 2A). Rainfall is highest in the northeastern and central parts of the basin, exceeding 0.008 m/day, particularly over the lake surface. Moderate precipitation

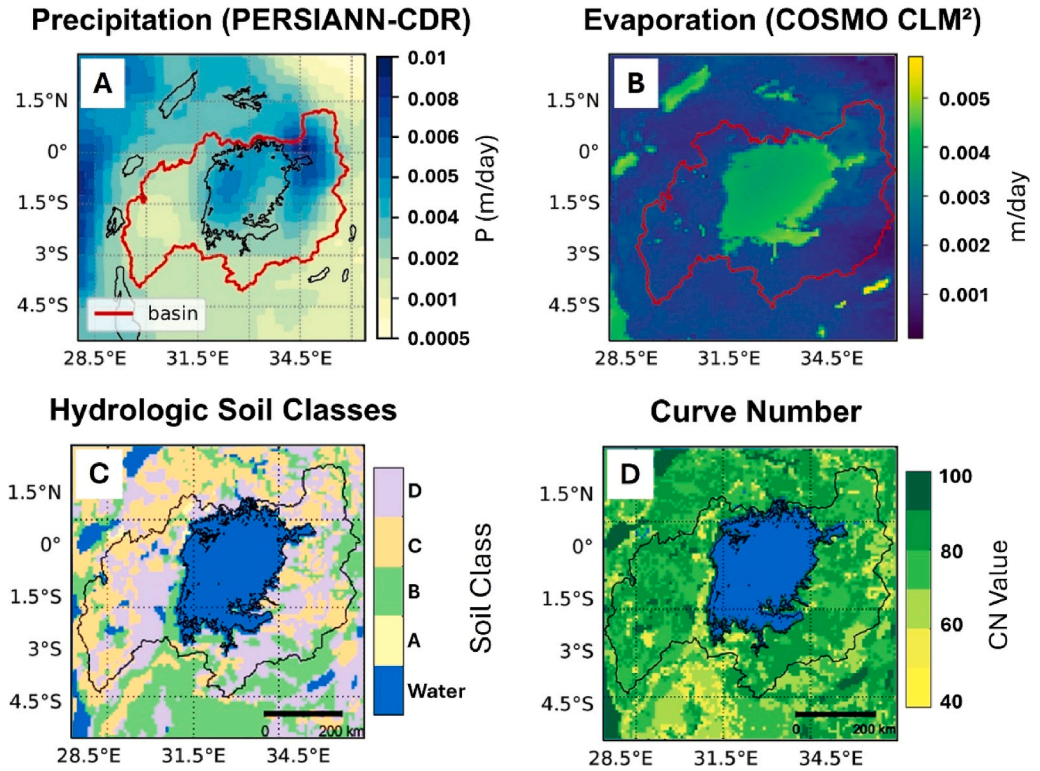


Fig. 2. for all $h_i \leq L(t)$Spatial distribution of key hydrological and environmental parameters across the study basin. **(A)** Mean daily precipitation derived from PERSIANN-CDR (m/day), highlighting spatial variability in rainfall intensity. **(B)** Mean daily evaporation rates modelled using COSMO-CLM^2 (m/day), indicating areas of higher atmospheric water loss. **(C)** Hydrologic soil classification based on infiltration potential, categorized into classes A (high infiltration) through D (low infiltration), with water bodies delineated. **(D)** Curve Number (CN) map representing runoff potential, where higher CN values correspond to reduced infiltration and increased surface runoff.

(0.003–0.005 m/day) dominates the western and southern regions, while the lowest values (around 0.001–0.003 m/day) occur toward the southeastern and southwestern fringes. The spatial pattern indicates a strong lake influence, with wetter conditions near the lake and drier zones outward, reflecting orographic and convective processes typical of equatorial climates.

Evaporation estimates derived from latent heat flux (LHF) using the COSMO-CLM^2 regional climate model with the rates being highest over the lake surface, reaching approximately 0.005 m/day, shown in bright yellow-green (Fig. 2B). Surrounding land areas exhibit lower rates, mostly below 0.002 m/day, indicated by darker blue shades. This spatial contrast highlights the lake's dominant role in regional water loss.

Hydrologic soil classification within the basin reveals a dominance of Class C soils, covering approximately 42.7% of the area, indicative of moderate infiltration potential (Fig. 2C). Class B soils account for 28.3%, supporting moderate water transmission. Class D, with low infiltration rates, comprises 18.9%, while highly permeable Class A soils occupy only 6.4%. Water bodies represent 3.7% of the mapped region. This distribution suggests a landscape prone to moderate runoff, with localized zones of high and low infiltration influencing hydrological response variability.

Curve Number maps indicate runoff potential and land surface hydrological response (Fig. 2D). High CN values (80–100) are observed in impervious and low-infiltration areas; open water surfaces (e.g., Lake Victoria) are assigned CN $\approx$ 100 but are excluded from runoff generation. While soil classes describe infiltration capacity, CN integrates both soil and land cover, and is therefore used as the primary runoff indicator. Regions like the Kagera and Nzoia basins likely experience higher runoff coefficients during intense rainfall due to steep gradients and reduced infiltration capacity, increasing flood susceptibility in these catchments. This spatial variability underscores the importance of terrain and land cover in hydrological response. (Fig. 2d).

The time series lake levels shown in Fig. (3) combines daily in situ observations from the WMO Hydrometeorological Survey (1948–1992) with satellite-derived measurements from the Database for Hydrological Time Series of Inland Waters (DAHITI) for 1993–2024. During the transition period (1993–1996), both datasets overlap and show strong agreement after bias correction and alignment to a common vertical datum, ensuring temporal consistency across the full record. The figure illustrates both short-term fluctuations and long-term trends in Lake Victoria's water levels. Between 1948 and the early 1960s, water levels remained relatively stable around 1134–1135 m a.s.l., followed by a sharp rise peaking in the mid-1960s. From the late 1960s through the 1980s, the lake experienced moderate variability, with intermittent peaks and declines. The 1990s onward show pronounced variability, including a notable drop in water levels around 2005–2006, followed by a steady recovery and peak levels observed in the 2020 s. This combined dataset highlights both natural hydrological variability and potential impacts from climatic and anthropogenic factors over the past seven decades.

4.2. *Evaluation of simulated lake levels*

A quantitative comparison between observed and simulated lake levels for Lake Victoria over the 1984–2023 period highlights the model's ability to reproduce both seasonal variability and decadal trends (Fig. 4). Key performance metrics, including the Nash–Sutcliffe Efficiency (NSE = 0.749), coefficient of determination (R^2 = 0.749), and Root Mean Square Error (RMSE = 0.262 m), affirm a robust model structure with high predictive skill. Furthermore, the LaVIWaB model successfully captures sharp hydrological events, such as the 1997–98 El Niño-induced rise and the significant post−2019 level increase, suggesting adequate sensitivity to climatic extremes and rainfall-runoff dynamics (Wang et al., 2025).

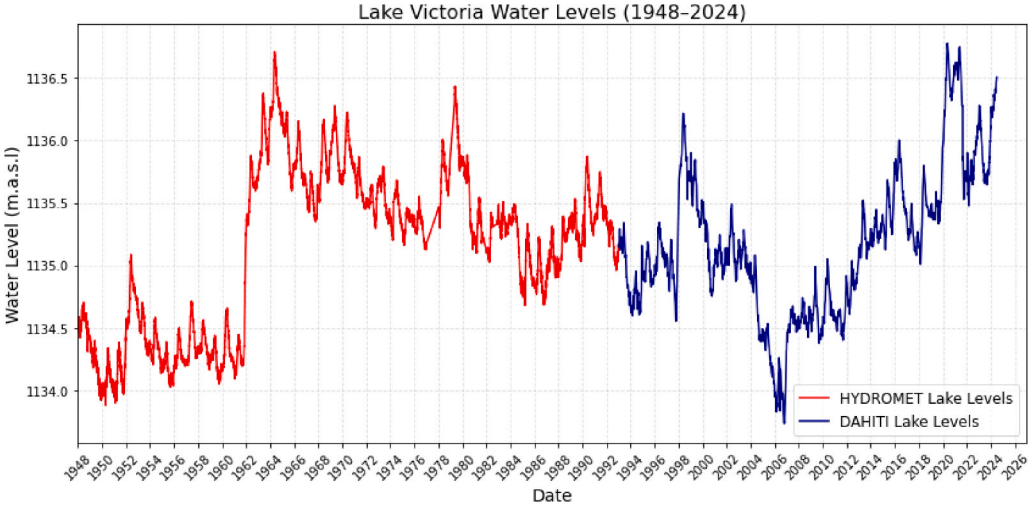


Fig. 3. Temporal variation in Lake Victoria's water levels from 1948 to 2024. The record integrates daily ground-based observations collected during the WMO Hydrometeorological Survey (1948–1992) with satellite-derived data from the Database for Hydrological Time Series of Inland Waters (DAHITI) for the period 1993–2024. DAHITI measurements, originally in meters above sea level (m a.s.l.), were shifted to align with in situ gauge references for consistency across the full time series.

However, the model underestimates some peak levels and overestimates troughs, particularly during the mid-2000s and the 2015–2019 period. These discrepancies likely reflect limitations in representing human-induced outflow regulation (e.g., dam operations deviating from the agreed curve) as discussed in Vanderkelen et al. (2018a), and unresolved groundwater–surface water exchanges. We note in this context that 2006–2011 is the period for which we replaced the Agreed Curve data with a regression-based analysis (see sect. 3.2.2), reflecting that the Agreed Curve was not a suitable approach for characterising lake outflows and lake levels in this period. The model does not fully reproduce the pronounced decline and subsequent rise in lake levels observed after 2020–2021, again likely due to deviations between actual dam releases and those prescribed by the Agreed Curve. During this period, unrecorded or emergency dam operations were undertaken to mitigate extreme inflows, which the model, based solely on the Agreed Curve, could not capture. The temporal coherence of the simulation post−2000 nonetheless highlights the benefits of improved input data and calibration. The smoother behaviour of the simulated hydrograph relative to observations further indicates a dampened response, which may arise from simplified assumptions and from the model's daily temporal resolution that gives sub-monthly regulation effects. Overall, we conclude that the LaVIWaB model is reasonably capable of representing the observed lake level dynamics, with remaining limitations likely linked to imperfect representation of outflow regulation as well as uncertainties in input forcing, particularly precipitation and evaporation.

4.3. Evaluation of simulated lake extent

Fig. 5 theoretically illustrates the progressive spatial inundation of Lake Victoria at discrete depth increments from 70 m to 90 m, based on a high-resolution bathymetric and topographic elevations (0 m water depth is at an elevation of 1052 m a.s.l). The panels reveal nonlinear shoreline expansion due to the lake's broad, low-gradient littoral zones. As depth increases, lake flooding propagates laterally, especially in the north and southeastern sub-basins, highlighting bathymetric and topographic controls on floodplain connectivity. These idealised depth-sliced masks serve as the basis for the subsequent water balance modelling, inundation forecasting, and simulating dynamic lake–wetland interfaces in climate-sensitive hydrological regimes.

The time series comparison between modelled, satellite-derived, and snapshot global validation datasets give consistency in modelled and observed lake area values (Fig. 6A). The different datasets; Copernicus Digital Elevation Model (2011: 66,829 km^2), Dataverse products (2015: 66,793 km^2), and Water Balance Model by Vanderkelen et al. (2018) (66,867 km^2) estimates align closely with the modelled lake extent values (66,835 km^2, 66,887 km^2, and 66,953 km^2, respectively), differing by less than 0.6% for each to the three snapshots. This consistency reinforces the representativeness of the LaVIWaB simulations across multi-source baselines. In contrast to the snapshot estimates, the comparison to the Wu et al. (2023a), (2023b) remote sensing product suggests an overestimation of the simulated lake area by around 400 km^2 (approximately 0.6%) (Fig. 6a). However, Moreover, Wu et al. (2023a), (2023b) report an average lake surface area of 66,473 km^2, whereas lake surface areas reported in the scientific literature range between 64,235 km^2 and 66,894 km^2 (Ali et al., 2024b; J. L. Awange et al., 2019; Nakkazi et al., 2024; Ogembo and Akinyi Mohamed, 2023; Pietroiusti et al., 2024; Thiery et al., 2015; Vanderkelen et al., 2018a; Wu et al., 2023a, 2023b). Given this and the close agreement between the WBM and LaVIWaB estimates (66,900 km^2 on average for the period 1984–2023), we suspect that the remote sensing product may be characterised by a small, systematic bias.

To address this systematic difference, we subsequently evaluate model behaviour by analysing anomalies. relative to the mean lake

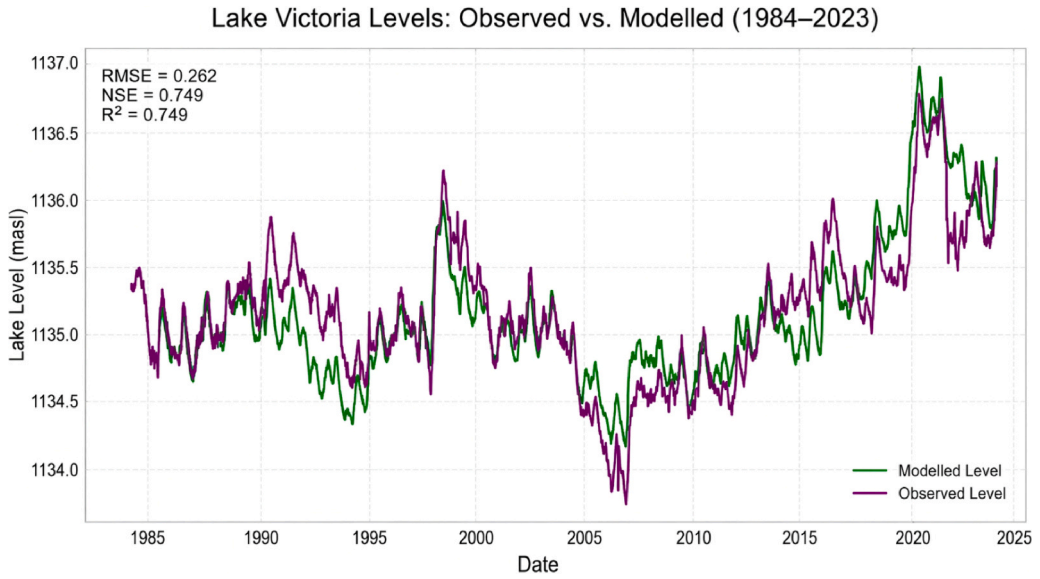


Fig. 4. Modelled and semi-observed lake levels for Lake Victoria from 1984 to 2023 (m a.s.l.). The observed elevation anomalies were not used for model calibration and are therefore independent from the modelled elevation anomalies.

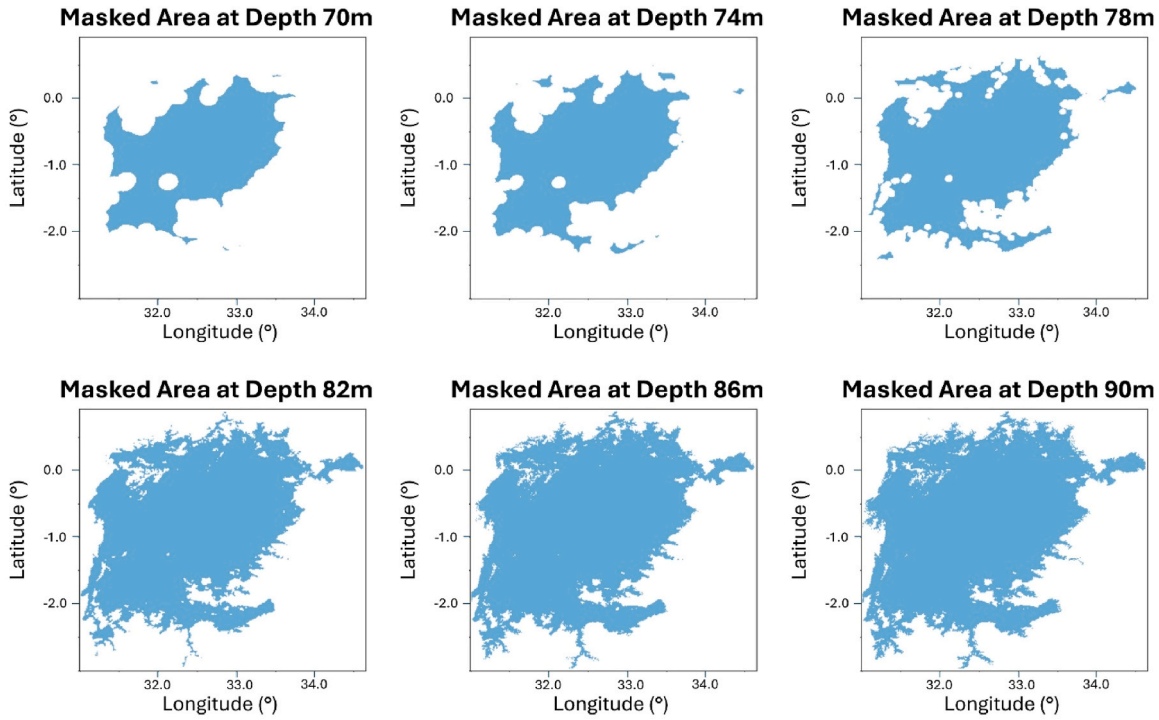


Fig. 5. Theoretical filling of Lake Victoria and surroundings at discrete maximum depth increments from 70 m to 90 m, based on lake area masked high-resolution bathymetry and DEM.

area over 2000–2023 from each product (Fig. 6B). The modelled series (solid blue line) closely follows the general behaviour of the satellite-observed estimates from Wu et al., (2023a), (2023b) (red dashed line), although the model consistently yields higher absolute area values, with an average offset of approximately 500 km^2. Both datasets indicate stable conditions from 2000 to around 2018, followed by a marked increase in lake area after 2019, consistent with the known hydrological rise in the lake during this period. The results show mildly synchronized variability between the LaVIWaB model and Wu et al., (2023a), (2023b) satellite data, particularly during the 2010–2022 hydrological transitions. However, the modelled anomalies suggest stronger interannual fluctuations in lake surface area relative to the satellite-based data, reflecting higher sensitivity to hydrological variability simulated by LaVIWaB. Despite marked differences in amplitude, both anomaly series exhibit similar directional changes during major hydrological events, particularly between 2019 and 2021. The anomaly Pearson's correlation between the Wu et al., (2023a), (2023b) dataset and the modelled series is $r = 0.598$, indicating a moderate relationship, while the Root Mean Square Error (RMSE) of 22.77 km^2 and Nash–Sutcliffe Efficiency (NSE) of 0.38 demonstrates reasonable agreement in magnitude. Although the RMSE of 22.77 km^2 may appear large in absolute terms, it represents less than 0.04% of the mean lake area (~66,900 km^2), indicating minor deviation relative to basin scale. The magnitude of this error is within the uncertainty range reported for optical remote sensing–based surface water products in persistently cloud-covered equatorial regions.

Overall, the results suggest that the LaVIWaB model reproduces observed lake area variability with moderate consistency, capturing key temporal patterns despite systematic differences in mean area estimates. Our results also highlight challenges for optical remote sensing-based products in capturing absolute lake areas values in tropical regions characterised by sustained high cloud cover. The integration of WBM-based validation enhances the robustness of the temporal assessment and confirms that the LaVIWaB framework maintains spatial coherence with global reference datasets (Messager et al., 2016; Yamazaki et al., 2019).

4.4. Flood simulations

Hypsographic curve of Lake Victoria shows the relationship between lake level elevation and lake surface area as estimated in the water balance model based on bathymetry of the lake and topography of the basin (Fig. 7). The results show a strong non-linear increase in area with rising lake levels. Notably, a significant inflection occurs around 1134 m a.s.l., beyond which area and volume expand less rapidly, reflecting the landscape's steep topography near the shores. For lake levels below the long-term average (i.e. lake drying) even modest water level changes result in large changes in area, reflecting the influence of gentle sloping lake beds near the lake shore. These curves underpin the critical importance of near-shore topography and bathymetry on lacustrine flooding and drying.

The modelled surface area of Lake Victoria over the entire simulation period (1984–2023) confirms the non-linear effect of shoreline topography by showing a step-like behaviour with several stable states (Fig. 8). The curve shows two main stable states of

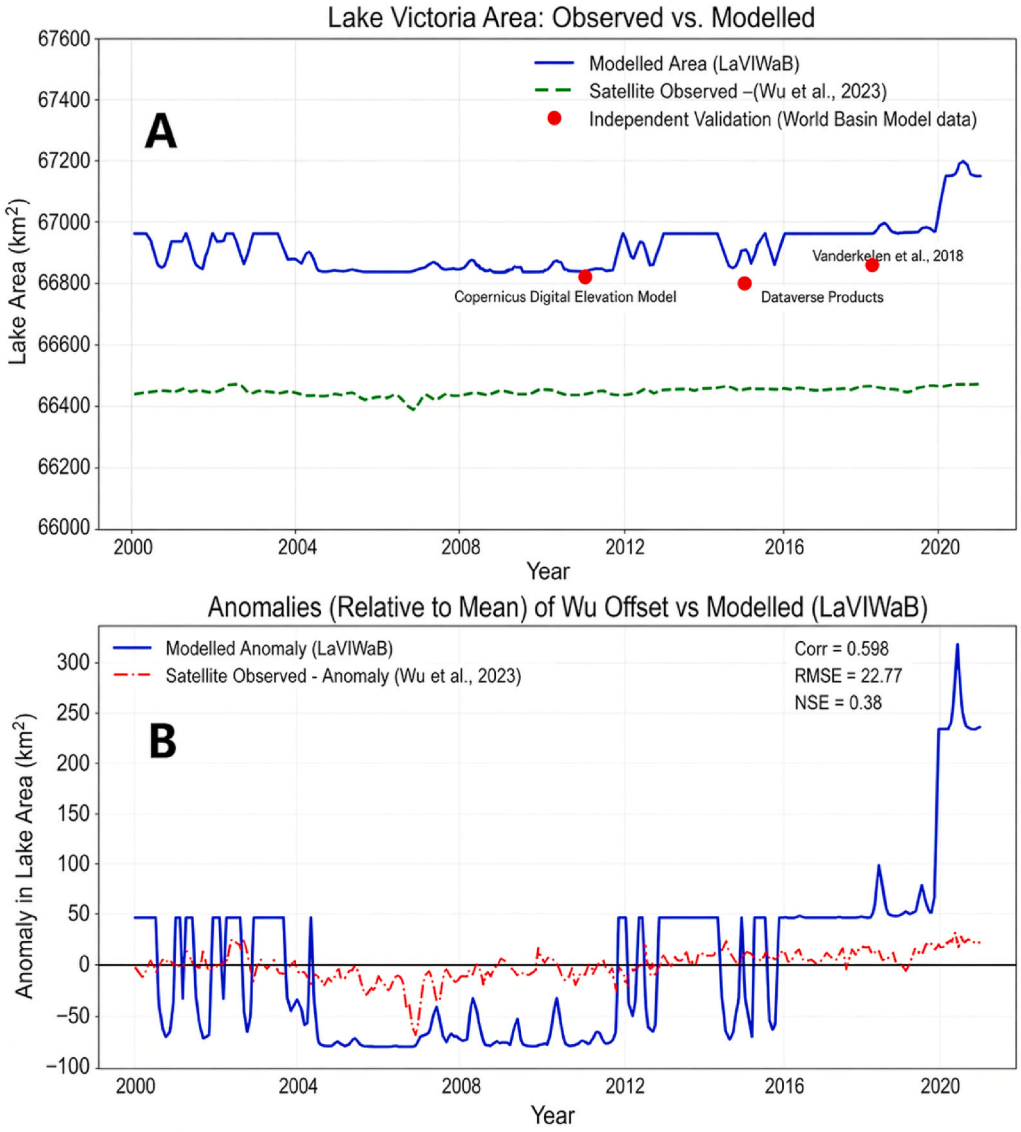


Fig. 6. Lake Victoria area as observed and modelled from 2000 to 2023. (A) The blue solid line represents the modelled lake area simulated using the LaVIWaB framework, while the green dashed line shows satellite-based observations from Wu et al. (2023a), (2023b). Red circular markers indicate independent validation points derived from different datasets; Copernicus Digital Elevation Model (2011: 66,829 km^2), Dataverse products (2015: 66,793 km^2), and Water Balance Model by Vanderkelen et al., (2018) (66,867 km^2). Note that the time stamp of the Vanderkelen et al. (2018a) lake mask is unclear and therefore assigned to the year of publication of the study. (B) Anomalies in lake area relative to the long-term mean, with the blue solid line showing modelled anomalies and the green dashed line showing satellite-derived anomalies offset as per Wu et al. (2023a), (2023b), confirming the robustness and temporal consistency of the LaVIWaB model.

66852 km^2 and 66967 km^2 for the period 1984–2019. However, from 2020 this regime shifts towards substantially higher values of around 67,158 km^2, during the record breaking lake area of 67,242 km^2 in May 17, 2020. The transition between both regimes is sharp and occurs around end of 2019. The apparent step-like behaviour in simulated lake area arises from the discretized elevation thresholds in the DEM–bathymetry dataset, where inundation expands only when lake levels exceed successive terrain elevations, resulting in a quantized area response consistent with the derived hypsometric relationship.

4.5. Mapping flood-prone areas (inundation zones) around the lake

Mapping of flood-prone areas (inundation zones) around Lake Victoria reveals a clear and spatially coherent pattern of shoreline vulnerability associated with extreme lake-level rise events. The elevation-based spatial impact analysis compares conditions during the relatively dry period of October 2018 with the record-breaking flooding of May 2020, during which Lake Victoria reached

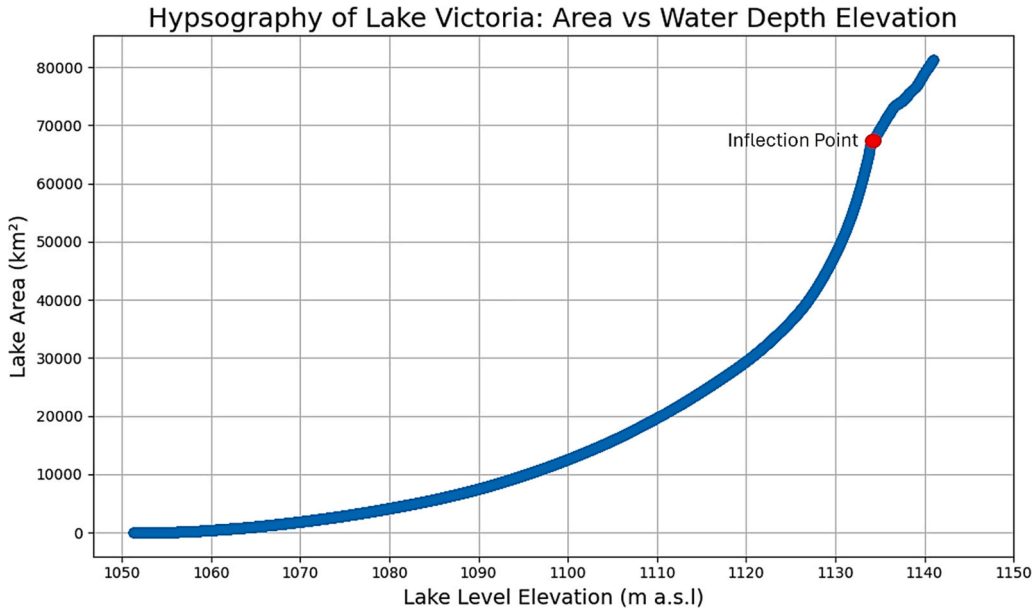


Fig. 7. DEM–bathymetry-derived hypsometric (elevation–area) relationship of Lake Victoria, showing the nonlinear relationship between lake level (m a.s.l.) and surface area used in the LaVIWaB model.The curve illustrates how Lake Victoria's surface area expands with increasing water level elevation and has an inflection point at 1134 m a.s.l. The lake bottom is at 1051.23 m a.s.l, with inflection point at 1134 m a.s.l and the historical maximum recorded lake level is 1136.77 m a.s.l**.** Beyond this maximum point, the curve represents projected area expansion for hypothetical scenarios where the lake level exceeds historical records.

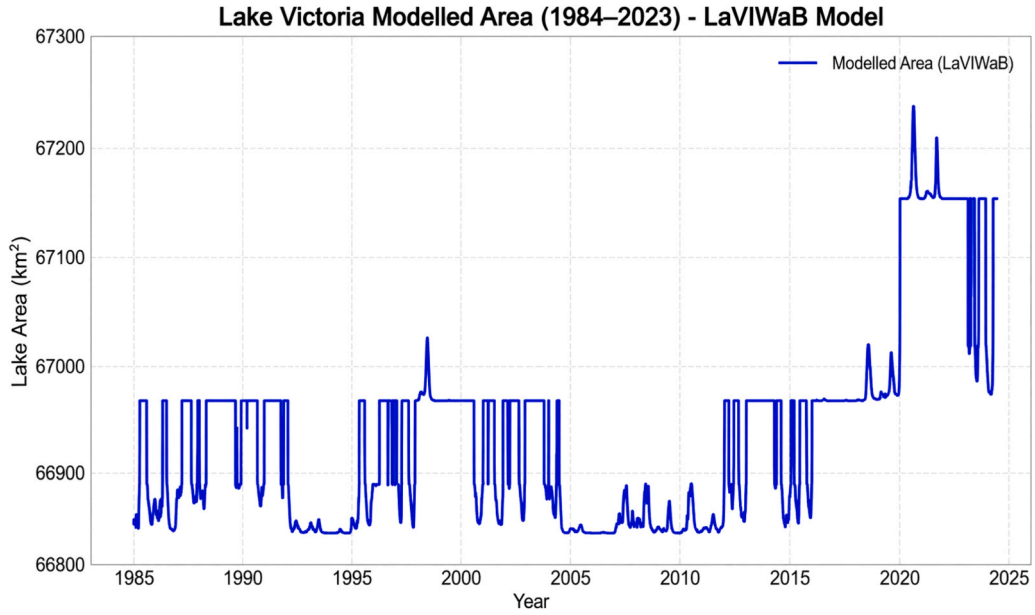


Fig. 8. Modelled surface area of Lake Victoria from 1984 to 2023 using the LaVIWaB framework.

historically high water levels exceeding 1134.5 m a.s.l. (Fig. 9). Areas highlighted in red represent locations that were inundated during May 2020 but remained exposed during October 2018, thereby delineating zones of episodic flooding driven by anomalous lake-level expansion. The results indicate that newly inundated areas are predominantly concentrated in low-lying terrain below approximately 1025 m a.s.l., corresponding to gently sloping littoral plains and deltaic environments surrounding the lake. These flood-prone zones are not uniformly distributed; instead, they exhibit pronounced spatial clustering in several key regions. In the northwestern sector, inundation is particularly evident along the Ugandan shoreline, including areas around Lwera wetlands, the Katonga River delta, and the Entebbe–Wakiso lowlands, where shallow gradients facilitate rapid lateral lake expansion during high-

water episodes. These zones are characterized by extensive wetlands and reclaimed land, which amplify flood exposure when lake levels rise even modestly.

In the southern and southeastern portions of the basin, flood-prone areas are concentrated along the Tanzanian shoreline, notably near Mwanza Gulf, Magu Bay, and the Simiyu River floodplain. These regions are geomorphologically defined by wide floodplains and low relief, making them particularly sensitive to lake-level anomalies. The inundation patterns observed here are consistent with backwater effects, where elevated lake levels impede river discharge and promote upstream flooding, thereby extending inundation well beyond the immediate shoreline. The northeastern shoreline, including parts of western Kenya such as Kisumu, Nyando River delta, Ahero floodplain, and the lower Nzoia River basin, also exhibits pronounced flood expansion. These areas are historically flood-prone due to their low elevation, high sedimentation rates, and strong coupling between river inflows and lake water levels. The May 2020 event exacerbated these dynamics, leading to widespread inundation of agricultural land, settlements, and infrastructure.

5. Discussion

This study introduces and evaluates the Lake Victoria Integrated Water Balance Model (LaVIWaB), a physically based modelling framework developed to simulate hydrological responses such as lake flooding under extreme climatic stressors in the Lake Victoria

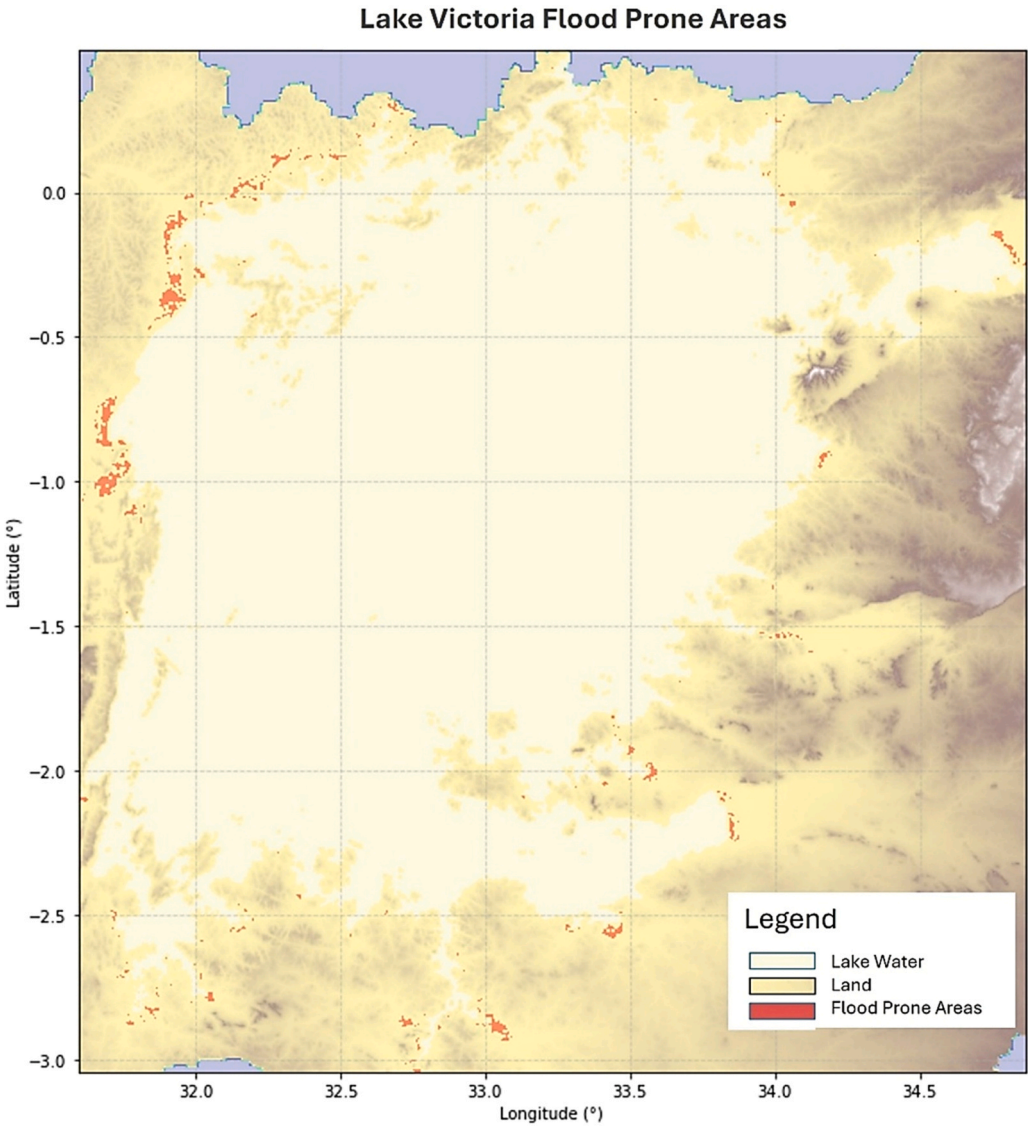


Fig. 9. Flood-prone areas of Lake Victoria identified by comparing modelled lake extents during October 2018 (low, 1135.10 m a.s.l.) and record-breaking flooding of May 2020 (high, 1136.77 m a.s.l.). Red zones around the lake indicate low-lying, high-risk areas for flooding and environmental stress.

Basin. Situated in one of the most hydro-climatically dynamic regions globally, Lake Victoria's flood regime poses significant threats to ecosystems, infrastructure, and the livelihoods of over 42 million people across Kenya, Uganda, and Tanzania (J. Awange, 2022). In this section we discuss model performance, physical insights, and implications for flood risk management and climate adaptation.

The results illustrated in Fig. 6 demonstrate the performance of the LaVIWaB model in reproducing the temporal variability of Lake Victoria's surface area between 2000 and 2022. The anomaly plot in Fig. 6B shows that LaVIWaB captures interannual variability more dynamically, with sharper peaks and troughs relative to the satellite record, reflecting greater sensitivity to hydrometeorological forcing. Despite amplitude differences, the anomaly trends are directionally consistent during major events. These findings confirm that LaVIWaB effectively represents temporal fluctuations in Lake Victoria's surface area, maintaining stability and consistency while accurately reflecting hydrological variability and major shifts in lake extent.

The reliability of the LaVIWaB flood inundation framework is supported by the calibration and validation procedures implemented in this study. Lake level simulations were evaluated against long-term observational records from WMO Hydromet and the DAHITI satellite altimetry dataset, while simulated lake surface area was compared against remote sensing products (Wu et al., 2023a, 2023b) and independent DEM-derived lake masks. These multi-source validation datasets provide complementary constraints on both vertical lake level dynamics and horizontal shoreline extent. The resulting performance metrics (NSE = 0.749; RMSE = 0.262 m for lake levels and RMSE = 22.77 km^2 for lake area anomalies) indicate that the model reproduces basin-scale hydrological behaviour with satisfactory accuracy for a large tropical lake system. However, uncertainties remain due to input data limitations, including satellite precipitation forcing, reanalysis-based evaporation estimates, and digital elevation model accuracy used for inundation mapping. These uncertainties propagate into simulated flood extent estimates, particularly in low-gradient littoral zones where small elevation errors can lead to large horizontal differences in inundation boundaries.

Previous studies that simulated Lake Victoria water balance using reanalysis and observational datasets provide a useful benchmark for evaluating the performance of the LaVIWaB framework. For example, the water balance modelling experiments of Vanderkelen et al. (2018a) reported lake-level simulation performance with NSE values between approximately 0.65 and 0.80 depending on the forcing datasets used, which is comparable to the performance obtained in this study (NSE = 0.749; RMSE = 0.262 m). Similarly, analyses using satellite altimetry and reanalysis forcing have reproduced interannual variability in Lake Victoria levels but highlighted substantial sensitivity to precipitation forcing datasets and evaporation estimates (Pietroiusti et al., 2024). The results fall within this performance range, indicating that the LaVIWaB model reproduces basin-scale lake dynamics with comparable accuracy to existing modelling frameworks.

The integration of WBM-derived validation points provides an independent benchmark for evaluating the credibility of LaVIWaB-simulated lake areas. The three validation datasets, generated through hydrologically consistent DEM conditioning and flow-routing algorithms, confirm the internal coherence of modelled outputs. The close alignment of simulated areas with water balance reference datasets for the years 2011, 2015 and 2018 substantiates the spatial accuracy of the LaVIWaB inundation module. Similar validation strategies have been employed in global hydrological studies, demonstrating that hybrid satellite–model frameworks yield superior performance for large lake systems (Messager et al., 2016; Sheng et al., 2016). Our evaluation approach not only improves confidence in long-term simulations but also establishes a reproducible template for integrating additional hydrological observations and future LaVIWaB model improvement efforts.

The core of the LaVIWaB model is its integrated treatment of hydrological processes such as precipitation, evaporation, tributary inflows, lake outflows, and storage changes, within a single mass balance framework. The model replicates the bimodal seasonal cycle characteristic of East African rainfall, with peak inflows and precipitation typically occurring in April-May and October-November (Nicholson, 2017b; Pietroiusti et al., 2024; Vanderkelen et al., 2018). This seasonal rhythm is a dominant control on Lake Victoria's hydrodynamics, as evidenced by the tight coupling between precipitation (P_{lake}) and riverine inflow (Q_{in}), and a slightly lagged response in outflow (Q_{out}), which reflects the lake's buffering capacity and regulated discharge via the Nalubaale Dam (Cook and Vizy, 2024; Pietroiusti et al., 2024; Vanderkelen et al., 2018). The model reveals that this seasonality is punctuated by interannual and decadal variability, most notably during major ENSO events such as 1997–98 and 2019–20. These periods correspond to extreme flooding episodes. The 1997–98 El Niño led to months of prolonged rainfall, pushing lake levels over historical averages and causing extensive flooding along the Kenyan and Ugandan shorelines (Nicholson, 2017). Similarly, between late 2019 and mid−2020, Lake Victoria rose to its highest recorded level since 1964, approximately 1136.77 m a.s.l., inundating settlements in Kisumu, Entebbe, and Mwanza (Shu et al., 2024). During that period, unprecedented rainfall drove lake levels up by approximately 1.21 m, displacing over 29,000 residents living within 50 km of the shoreline (Pietroiusti et al., 2024). LaVIWaB successfully replicates such anomalies, revealing complex seasonal and interannual variability, including the tight coupling between precipitation peaks in April–May and October–November, and delayed outflows modulated by government-controlled dams.

The LaVIWaB model accurately captures these anomalies, suggesting robustness in responding to large-scale climate variability. It also identifies residual fluxes during periods of hydrological imbalance, in which the model performs less reliably, indicating the influence of additional components like groundwater exchange and unregulated water withdrawals. These findings are consistent with the work of Taye et al. (2011) and Vanderkelen et al. (2018a), who highlight the importance of including human-hydrological interactions, such as irrigation demand and reservoir regulation, to achieve basin-wide water balance closure. Another critical dimension is the role of regional land-use change, including deforestation, wetland degradation, and urban sprawl. These changes intensify surface runoff and alter evapotranspiration patterns, thus contributing to peak flows that amplify flood risk. The lake extent trend compared with changing land cover is evident in simulations post−2010, where intensifying runoff signatures match rapid urbanization trends in cities like Kisumu and Kampala (Ogembo et al., 2022). Incorporating dynamic land cover data into future iterations could further enhance model accuracy and responsiveness.

An innovative feature of this study is the integration of lake level dynamics with surface area estimates derived from high-

resolution bathymetry and topography data. Unlike linear lake response models, LaVIWaB employs an elevation–area–volume relationship that reflects the nonlinear nature of lake expansion and contraction. This is particularly important for Lake Victoria due to its marked contrast between broad littoral zones with low topographic gradients and steep onshore topography (Figure A2). Another important insight relates to the differential rainfall-runoff response of various sub-basins within the Lake Victoria watershed (Fig. 2c, sect 4.1). This heterogeneity underscores the need for spatially disaggregated catchment runoff modelling to incorporate sub-catchment responses and lag times and improve the realism of downstream lake level simulations.

The analysis linking lake level to surface area in Fig. 7 underscores a threshold-like behaviour of the system at high lake levels. The curve steepens sharply between 1132 and 1134.5 m a.s.l., after which it gives a gentle uneven slope due to the varying topography at the lake shores and the much steeper topography further away. The flatter terrain close to the shores means that even minor increases in water level translate into disproportionately large expansions of flooded surface area. To ensure these values are meaningful to stakeholders, technical lake-level measurements must be expressed in terms that are both standardized and actionable. In the case of Lake Victoria, dam operators and water managers often rely on the Jinja Pier gauge as the primary reference for water-level decisions (Khaki and Awange, 2021; Sutcliffe and Petersen, 2007). These gauge readings, recorded in meters, are routinely converted into absolute elevation above mean sea level (m a.s.l.) using reference datums such as the Khartoum Datum. For example, the highest lake level ever recorded in the 20th Century was 13.46 m on the Jinja Pier gauge, corresponding to 1135.11 m a.s.l. (Khartoum Datum) or 1136.35 m a.s.l. (alternative datum) (Kite, 1981). This calibration facilitates consistency across hydrological modelling, dam management, and regional flood forecasting systems.

To translate these findings into actionable guidance, we propose practical threshold levels for water users based on observed hydrological extremes and modelled lake–area relationships. The record-breaking 2020 flood event reached 1136.77 m a.s.l., inundating extensive low-lying areas (Fig. 9), while relatively stable long-term levels before the 1960s were around 1134–1135 m a.s.l. (Fig. 3). Our hypsographic analysis (Fig. 7) indicates that the system begins to respond non-linearly once lake levels surpass 1132 m a.s.l., with small increases translating into disproportionately large expansions of flooded area. Accordingly, we define three actionable thresholds: “Alert” at 1134–1135 m a.s.l., where early preparedness measures should be triggered; “Danger” at 1135–1135.5 m a.s.l., where critical infrastructure and vulnerable communities along the shoreline face elevated flood risks; and “Critical” at $\geq$ 1136 m a.s.l., corresponding to record-breaking conditions such as those observed in 2020 and 2024, where widespread displacement and economic losses are highly probable. By linking gauge readings (e.g., Jinja Pier) to absolute elevations above sea level, these thresholds provide a scientifically robust yet easily interpretable framework for dam operators, policymakers, and local communities. Establishing such operational benchmarks would significantly strengthen flood early warning systems and support proactive adaptation planning in the Lake Victoria Basin (Ogembo et al., 2026).

Despite its strengths, some limitations constrain the current version of LaVIWaB. First, satellite-derived lake area estimates suffer from issues such as cloud interference, seasonal vegetation cover, and sensor resolution limitations. These factors are exacerbated in earlier datasets, particularly before the launch of Sentinel−2 in 2015, meaning the validation data itself presents uncertainties. Future work should therefore evaluate inundation sensitivity using next-generation terrain datasets such as FABDEM or Copernicus GLO−30, which reduce vegetation and canopy biases present in earlier DEM products. Second, inaccuracies in digital elevation models (e.g., ASTER GDEM) introduce errors in inundation boundaries as computed by the model, especially in gently sloping terrains where vertical errors of < 1 m can result in large horizontal discrepancies as well as its relatively course resolution. Third, LaVIWaB currently employs a static inundation algorithm, which does not dynamically adjust for changing vegetation, land use, sedimentation, or flood protection measures (e.g. levees). Furthermore, anthropogenic control via dam operations introduces additional modelling challenges. For instance, between 2006 and 2011, deviations from the "Agreed Curve" release policy at Nalubaale Dam resulted in artificial fluctuations in lake level (Kull, 2006). The model's limited capacity to simulate these regulated outflows suggests the need to incorporate dam operation modules.

Another limitation arises from the lack of real-time data assimilation, particularly for variables such as groundwater recharge, inter-basin transfers, and human abstraction. The growing application of remote sensing platforms like SWOT, GRACE, SMAP, and DAHITI could offer valuable inputs for model calibration (Chen et al., 2022; Mladenova et al., 2019; Yao et al., 2025). Integrating these into a real-time hydrological dashboard could significantly improve operational flood forecasting and water resource management. For instance, SWOT offers great potential lake extent data but since it was recently launched, it currently provides insufficient time series data for this study.

The LaVIWaB model provides valuable spatial information for flood risk management. Flood extent maps generated for the 2020 flood event reveal hotspots of recurrent inundation below 1125 m a.s.l., particularly in the western basin (Busia, Siaya), southern margins (Mwanza), and northeastern deltas (Napak, Uganda). These regions coincide with high population densities and agricultural productivity, making them vulnerable to both economic losses and humanitarian crises. The utility of LaVIWaB extends beyond retrospective analysis. Its capability to simulate future lake levels and inundation scenarios under different rainfall and management conditions positions it as a tool for climate adaptation planning (Vanderkelen et al., 2018b). For example, under high emissions scenarios, such as RCP8.5, East Africa is projected to receive more intense and less predictable rainfall, raising the probability of extreme flood events (Ayugi et al., 2022; Souverijns et al., 2016; Thiery et al., 2016). With scenario-based inputs, LaVIWaB can be used to evaluate the efficacy of flood defences, inform insurance schemes, and prioritize investments in nature-based solutions such as wetland buffers and upstream reforestation (Lallemant et al., 2021).

Importantly, the model supports the implementation of early warning systems by providing probabilistic forecasts of lake level trajectories under various scenarios. In partnership with regional agencies managing the basin, LaVIWaB could serve as a back-end simulation engine for community-based disaster risk reduction initiatives. Furthermore, it could enhance anticipatory humanitarian responses through better resource pre-positioning and early evacuation protocols. Moreover, the transboundary nature of Lake

Victoria makes cooperative water governance imperative. The lake feeds the White Nile, and its outflow regulation directly influences downstream hydro-politics. LaVIWaB's capacity to simulate lake responses under varied climatic and management conditions can support the development of basin-wide treaties, early warning systems, and joint disaster response protocols. It aligns with the objectives of the Nile Basin Initiative and the Lake Victoria Basin Commission, offering a scientific foundation for shared water resource management and conflict avoidance.

Finally, the platform's scalability makes it potentially suitable for application in other African Great Lakes, including Lake Tanganyika and Lake Albert, where similar hydroclimatic dynamics and flood challenges persist. Future research should focus on improving representation of anthropogenic controls and hydrological feedbacks within the modelling framework. In particular, incorporating dynamic dam operation rules for the Nalubaale–Kiira complex would improve simulation of regulated outflows during extreme hydrological events. With further development, LaVIWaB could be repurposed to model urban flooding in rapidly growing lakeside cities or to support sediment transport studies and fisheries management. This versatility enhances its long-term value to both scientific and policy communities.

6. Conclusion

This study developed and applied the Lake Victoria Integrated Water Balance (LaVIWaB) model to simulate lake level dynamics and shoreline flooding in the Lake Victoria Basin. The model addresses limitations in previous Lake Victoria water balance models that assumed constant lake surface area by coupling hydrological mass balance with a dynamic elevation–area relationship derived from high-resolution bathymetric and topographic datasets. This integration allows the model to translate hydrological fluxes directly into spatially explicit inundation dynamics, enabling improved analysis of flood behaviour in large tropical lake systems.

Model evaluation demonstrates that LaVIWaB reproduces observed lake level variability with strong performance (NSE = 0.749; RMSE = 0.262 m) and captures major hydrological anomalies such as the 1997–1998 El Niño and the extreme 2019–2020 flooding event. Simulations reveal a pronounced nonlinear relationship between lake level and surface area expansion, with critical thresholds emerging once water levels exceed approximately 1132–1134 m a.s.l. At these elevations, even small increases in water level can trigger substantial lateral expansion of flooded areas due to the gentle slopes of Lake Victoria's littoral zones. These dynamics highlight the vulnerability of low-lying shoreline regions and provide important insights for flood risk management in the basin.

Several methodological limitations remain. Uncertainty arises from satellite-derived precipitation forcing, evaporation estimates, and digital elevation model accuracy used in inundation mapping. In addition, simplified representation of groundwater interactions and dam operations may affect the precision of simulated lake responses during extreme hydrological events. Future work should focus on improving representation of anthropogenic regulation and hydrological feedbacks within the modelling framework. Integration of emerging satellite observations (e.g., SWOT, GRACE) would further enhance model calibration and support near-real-time flood monitoring. Overall, the LaVIWaB model provides a scalable and physically consistent framework for analysing flood dynamics in the Lake Victoria Basin and offers a useful tool for flood early warning, climate adaptation planning, and transboundary water management.

CRediT authorship contribution statement

Wim Thiery: Writing – review & editing, Supervision, Methodology. **Gavin Akinyi:** Visualization, Project administration, Formal analysis. **Mary Akurut:** Validation, Data curation. **Inne Vanderkelen:** Validation, Methodology. **Rosa Pietroiusti:** Writing – review & editing, Software, Methodology. **Vincent Ogembo:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author(s) used ChatGPT (OpenAI) to assist with language editing, improvement of clarity, and structuring of the text. After using this tool, the author(s) critically reviewed, revised, and edited all content and take full responsibility for the accuracy, originality, and integrity of the published work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code availability

The LaVIWaB (Lake Victoria Integrated Water Balance) Model code used in this study is available here: https://github.com/VINCENTOGEMBO/LaVIWaB/. The code was developed in Python and includes modules for hydrological simulation, bathymetric processing, and flood extent mapping.

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Appendix 1. Bathymetric plan view of Lake Victoria

The map illustrates the spatial variation in lakebed depth (in meters) across Lake Victoria, derived from the General Bathymetric Chart of the Oceans (GEBCO) dataset (Figure A1). Deeper regions are concentrated in the central basin, shown in yellow-green, while shallower areas along the littoral zones are indicated by darker shades. Depth gradients reflect the lake's topographic complexity and are essential for hydrodynamic modelling, lake volume estimation, and flood simulation analysis.

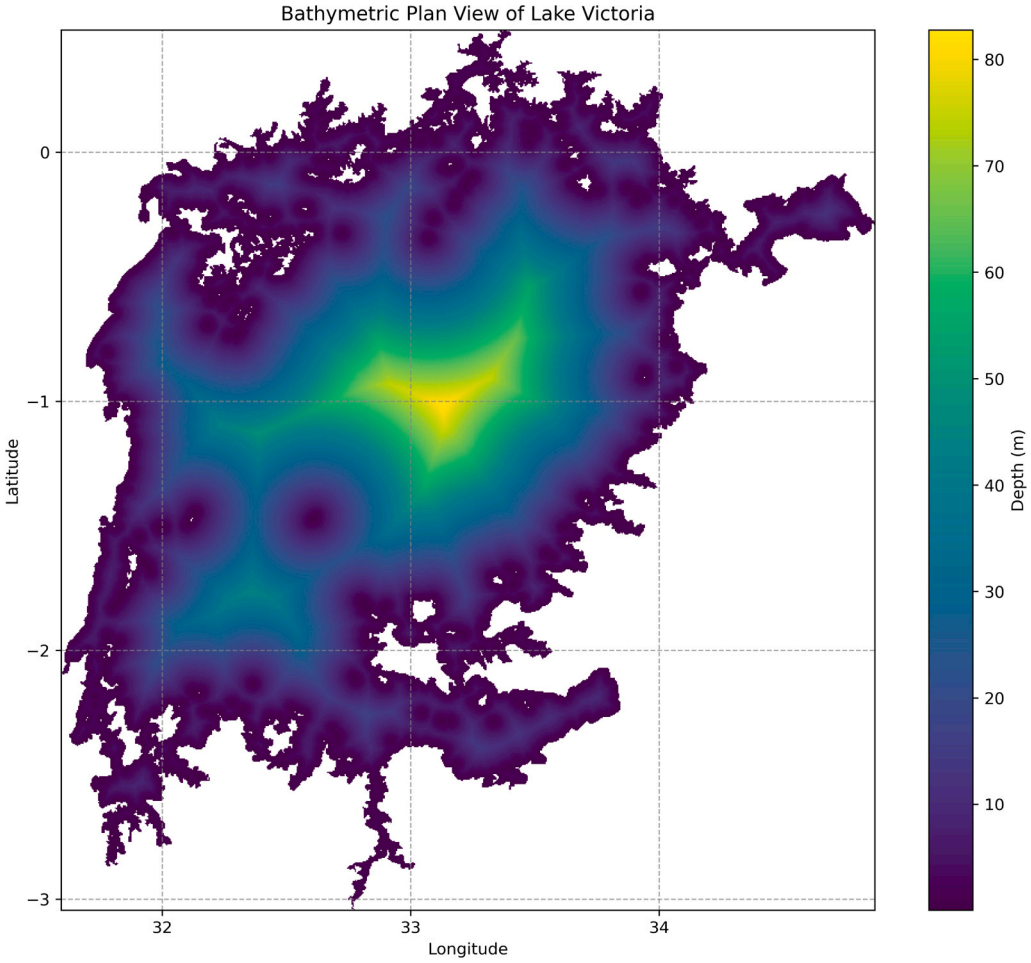


Figure A1. Bathymetry plan view of Lake Victoria

Appendix 2. Bathymetric and topographic profile of Lake Victoria and its basin

The bathymetry and topography of Lake and basin were interpolated and their profile extracted along latitude −0.8° (Figure A2). The transect integrates Shuttle Radar Topography Mission (SRTM) elevation data and GEBCO-derived bathymetry to visualize the

transition between terrestrial and subaqueous zones. The basin exhibits a wide and gently sloping lake floor, with maximum depths near the central portion and steep escarpments on both the eastern and western margins. This morphological configuration explains the pronounced lateral inundation response to relatively minor water level changes, as small vertical fluctuations translate into moderate horizontal shoreline displacements. The derived profile supports the hypsometric modelling and lake surface area–elevation relationships used in the LaVIWaB model to simulate dynamic inundation and water storage variations.

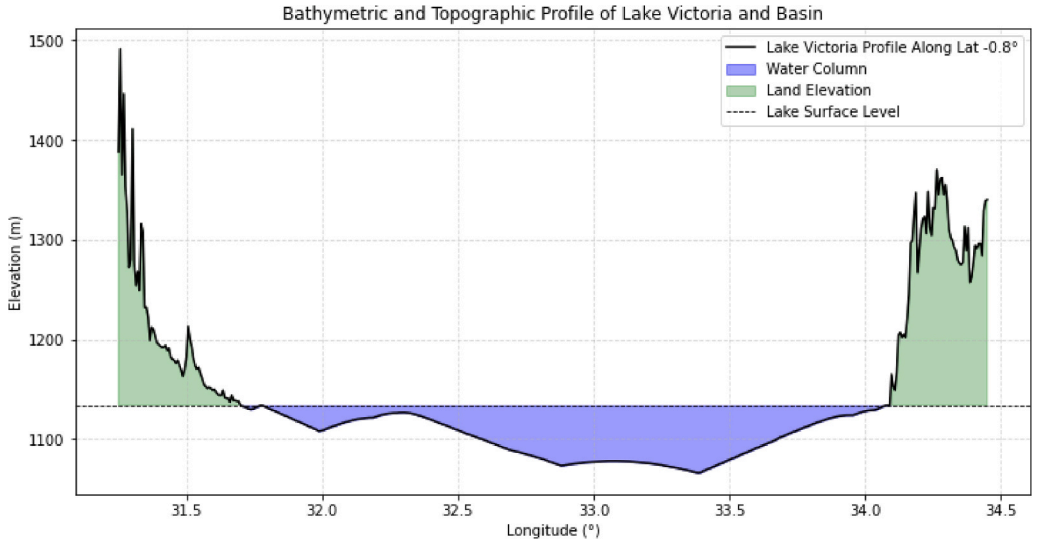


Figure A2. Bathymetric and Topographic Profile of Lake Victoria and its basin along 0.8°S

Appendix 3. Reconstruction of Lake Outflow for the 2006–2011 Period

The Lake outflow Agreed Curve contained discrepancies for the period between 5 March 2006 and 31 December 2011 due to regulations of the lake Victoria outflow at Jinja dam, Uganda. To address this, the affected interval was isolated, and a regression model was developed using the relationship between lake level and outflow from the surrounding valid periods. The model was then applied to estimate outflow values for the missing timeframe (Figure A3).

A power-law model was fitted:

$$Q_{out} = a(L - L_0)^b$$

where Q_{out} is outflow ($m^3\ s^{-1}$), L is lake level (m a.s.l.), L_0 is a reference level, and a, bare fitted parameters. Calibration used observed data from pre−2006 and post−2011 periods. The regression demonstrates strong performance, with high explanatory power ($R^2 = 0.93$), moderate error magnitude (RMSE = 251 $m^3\ s^{-1}$), and minimal positive bias (MBE = +19 $m^3\ s^{-1}$), indicating reliable outflow reconstruction.

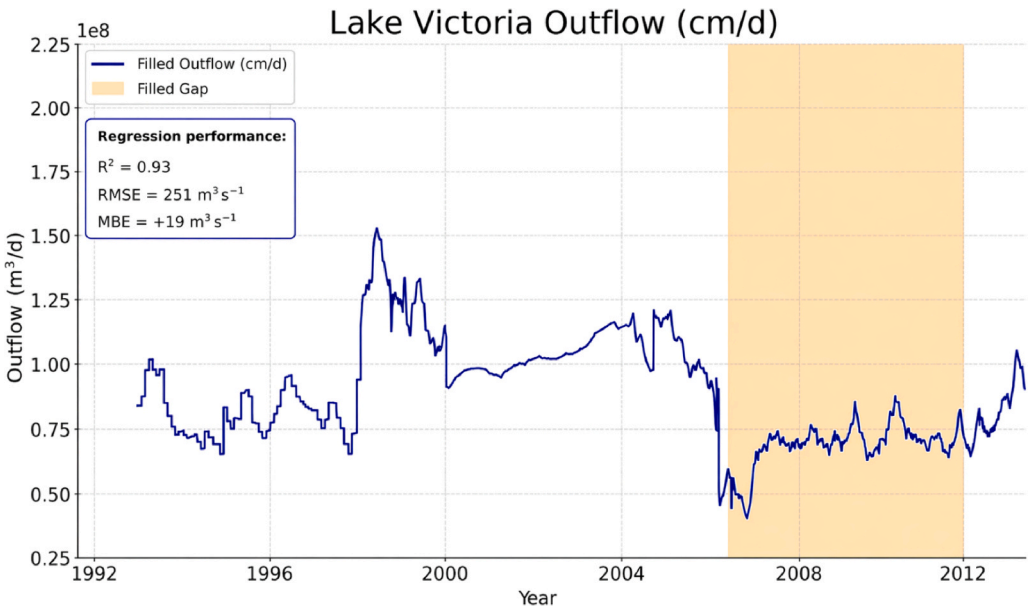


Figure A3. Isolated missing data processed by regression method

Data availability

Data will be made available on request.

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